

# Timothy Gao

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## EDUCATION

### University of California, Berkeley

Bachelor of Science in Electrical Engineering and Computer Sciences, 3.82 GPA

Berkeley, CA

May 2027

## EXPERIENCE

### Etched

Inference Intern

May 2026 - Current

San Jose, CA

- Writing custom attention and MOE kernels for GPT-OSS 120B inference (TP2EP4) for Etched's in-house silicon

### Nvidia

AI Developer Technology Intern

Aug. 2025 - Dec. 2025

Santa Clara, CA

- Developed mixed-precision **disaggregated LLM serving** in TensorRT-LLM by introducing a novel transfer-datatype feature in the KV-cache transceiver, enabling automatic precision conversion between prefill and decode
- Optimized end-to-end KV-cache transfer for disaggregated serving in **TensorRT-LLM**, spanning Python/C++ orchestration, NIXL transport, and CUDA kernels, reducing end-to-end latency by up to **26%** across batch sizes via FP16 prefill/FP8 decode while preserving full-precision baseline MMLU and GSM8K accuracy

### Robotic AI and Learning Lab

Undergraduate Researcher

Jan. 2025 - Present

Berkeley, CA

- Co-developed a hierarchical driving policy leveraging VLM priors to reason about long-tail scenarios and steer a VLA model via language conditioning, outperformed **SOTA** on Bench2Drive with biggest gains on Bench2Drive-LongTail
- Published *SteerVLA: Steering Vision-Language-Action Models Toward Effective Long-Tail Driving* under **Sergey Levine and Chelsea Finn**, presented at **RSS**, currently researching scalable off-policy methods in RL

### Jane Street

First-Year Trading and Technology Program

March 2024

New York, NY

- Created an autonomous **high-frequency trading bot** employing statistics, arbitrage, hedging, pennyng, and market making algorithms to trade bonds, stocks, ETFs, and ADRs, **placed 4th** in internal Electronic Trading Challenge

### Berkeley Artificial Intelligence Research, Trevor Darrell Lab

Undergraduate Researcher

Jun. 2024 – Aug. 2024

Berkeley, CA

- Developed and dockerized vision transformer model to detect military vehicles in satellite imagery for the NGA
- Implemented data processing pipeline to clean, chip, augment, and class-balance over **100k** raw satellite imagery
- Designed ScaleDet model using **PyTorch** and **MMDetection** with ScaleMAE Backbone and RCNN detection head, that encodes satellite metadata (e.g., ground sample distance) into feature embeddings
- Pretrained on Imagenet and xView, achieved **.17 mAP** after fine-tuning on NGA dataset and surpassing baselines

### K-Scale Labs

Machine Learning Engineer Intern

May 2024 — Aug. 2024

Palo Alto, CA

- Fine-tuned ViT-based image **depth estimation** model (DepthAnything) on in-house stereo data for real-time **3D perception** on humanoid platforms, achieving +26% AbsRel vs ResNet baseline in indoor/outdoor scenes, and collaborated to integrate it into K-OS, Stompy's custom on-board perception stack
- Fine-tuned vision-language-action model using **LoRa** for 7-DoF **robotic arm manipulation** in Pick and Place tasks, using data generated with PPO in simulation environment and achieving over 40% success rate

### MIT BWSI Summer Institute

Student

Apr. 2022 – Nov. 2022

Cambridge, MA

- Optimized image object detection using QHED, achieving  $O(\log^2 N)$ , exponential speedup over naive Sobel filter
- Wrote paper and implemented with qubit circuit blocks in **Qiskit SDK**, presented at Lincoln Labs symposium

## HONORS AND AWARDS, AFFILIATIONS, COURSEWORK

**Honors & Awards:** ICPC North America Championship Qualifier (Berkeley representative), **TreeHacks** 11.0 Grand Prize Winner & Tesla AI Excellence Prize, Cal Hacks 10.0 **Best Hack on AI**, D. E. Shaw Nexus Fellow, MIT Battlecode 2nd place, 3x USACO Platinum Qualifier, AIME Qualifier, **Codeforces** Master

**Affiliations:** CS 189 TA (Intro to ML Sp26), Machine Learning @ Berkeley, Ballers FC (IM Soccer Team Captain)

**Coursework:** Theoretical Statistics, Fourier Analysis, Complex Analysis, Information Theory, Random Processes, Operating Systems, Compilers, Computer Architecture, Linear Algebra, Algorithms, Databases, Security, CV, NLP, RL